

Advanced Data Computing (405 @ LU) Notes

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Welcome

Part I

SQL

1 Accessing Databases and Database Tables

1.1 resources

- [R for Data Science \(2e\) - Chapter 21](#)
- [Modern Data Science with R \(3e\) - Chapters 15 and 16](#)
- [Notes from Database Systems at UC Berkeley](#)
- [Introduction to dbplyr](#)
- [Hands-on Database \(2e\) - Steve Conger](#)
- CMSC 225 - Database Techniques and Modeling (F'26 and S'27)

1.2 load packages

```
library(tidyverse)
library(mdsr)
library(DBI)
library(RMariaDB)
library(dbplyr)
```

1.3 databases and accessing them

```
# access airlines database

con_air <- mdsr::dbConnect_scidb("airlines")
```

```
# access wai database

con_wai <- DBI::dbConnect(
  RMariaDB::MariaDB(),
  dbname = "wai",
  host = "scidb.smith.edu",
  user = "waiuser",
  password = "smith_waiDB")
```

1.4 database tables and accessing them

```
# lists tables in the database
```

```
DBI::dbListTables(con_air)
```

```
--- lists tables in the database
```

```
SHOW TABLES;
```

```
# access flights table from the airlines database
```

```
flights <- dplyr::tbl(con_air, "flights")
```

```
DESCRIBE flights;
```

```
# access carriers table from the airlines database
```

```
carriers <- dplyr::tbl(con_air, "carriers")
```

```
DESCRIBE carriers;
```

```
# carriers lives in the SQL database and is linked remotely
```

```
carriers |>
  utils::object.size() |>
  print(units = "Kb")
```

```
# nycflights13::flights |>
```



```
# object.size() |>  
# print(units = "Mb")
```

```
# carriers lives in R
```

```
carriers |>  
  dplyr::collect() |>  
  utils::object.size() |>  
  print(units = "Kb")
```

```
carries_df <- carriers |> dplyr::collect()
```

```
# terminate the SQL connection
```

```
DBI::dbDisconnect(con_air, shutdown = TRUE)
```

2 Writing SQL queries

2.1 load packages and connect to database server

```
library(tidyverse)
library(mdsr)
library(DBI)
library(RMariaDB)
library(dbplyr)
```

```
# access airlines database
```

```
con_air <- mdsr::dbConnect_scidb("airlines")
```

```
DBI::dbListTables(con_air) # list tables in database
```

```
# access flights table from the airlines database
```

```
flights <- dbplyr::tbl(con_air, "flights")
```

```
DESCRIBE flights;
```

2.2 SQL in R: Part 1 - the 'dbplyr' R package will directly translate 'dplyr' code into SQL

```
yrs <- flights |>
  summarize(min_year = min(year, na.rm = TRUE),
            max_year = max(year, na.rm = TRUE))

yrs
```

```
dplyr::show_query(yrs)
```

```
la_bos <- flights |>
  filter(year == 2013 &
    ((origin == "LAX" & dest == "BOS") |
     (origin == "BOS" & dest == "LAX")))

la_bos
```

```
dplyr::show_query(la_bos)
```

```
la_bos <- la_bos |>
  collect()

la_bos
```

```
# terminate the SQL connection

DBI::dbDisconnect(con_air, shutdown = TRUE)
```

2.3 SQL in R: Part 2 - using the DBI R package to write SQL queries in r code chunks

```
DBI::dbGetQuery(con_air,
  "SELECT * FROM flights LIMIT 8;")
```

```
DBI::dbGetQuery(con_air,
  "SELECT year, count(*) AS num_flights FROM flights GROUP BY year ORDER BY num_flights;")
```

```
# terminate the SQL connection

DBI::dbDisconnect(con_air, shutdown = TRUE)
```

2.4 SQL in R: Part 3 - writing SQL queries in sql code chunks

```
SELECT MIN(year) AS min_year,  
       MAX(year) AS max_year  
FROM flights;
```

```
SELECT *  
FROM flights  
WHERE (year = 2013 AND  
      ((origin = 'LAX' AND dest = 'BOS') OR  
       (origin = 'BOS' AND dest = 'LAX')));
```

```
SELECT *  
FROM flights  
LIMIT 2;
```

```
SELECT year,  
       COUNT(*) AS num_flights  
FROM flights  
GROUP BY year  
ORDER BY num_flights;
```

Queries in SQL start with the **SELECT** keyword and consist of several clauses, which have to be written in this order:

- **SELECT** allows you to list the columns, or functions operating on columns, that you want to retrieve.
- **FROM** specifies the table where the data are.
- **JOIN** allows you to stitch together two or more tables using a key.
- **WHERE** allows you to extract rows according to some criteria.
- **GROUP BY** allows you to aggregate the records according to some shared value.
- **HAVING** is like a WHERE clause that operates on the result set—not the original rows themselves.
- **ORDER BY** orders the rows of the result set.
- **LIMIT** restricts the number of rows in the output.

```
DESCRIBE flights;
```

```

-- display the first ten rows of the flights table

-- display year, month, day, origin, dep_time, sched_dep_time, dep_delay for the first ten rows

-- count the total number of rows in the flights table

-- extract all info about flights that flew from Appleton (ATW) to Chicago (ORD) in 2013

-- count the number of flights from Appleton (ATW) for each year

-- which carriers flew from Appleton (ATW) to Chicago (ORD)
-- how many flights did they do per year

-- for flights that flew from Appleton (ATW) to Chicago (ORD)
-- what was the latest flight for each carrier

-- for flights that flew from Appleton (ATW)
-- which destination had the lowest average arrival delay time

-- for flights that flew from Appleton (ATW)
-- consider only destinations with at least 2000 flights from Appleton (ATW)
-- which destination had the lowest average arrival delay time

# terminate the SQL connection

DBI::dbDisconnect(con_air, shutdown = TRUE)

```

3 Joins, Unions, and Subqueries

3.1 load packages and connect to database server

```
library(tidyverse)
library(mdsr)
library(DBI)
library(RMariaDB)
library(dbplyr)
```

```
# access airlines database
```

```
con_air <- mdsr::dbConnect_scidb("airlines")
```

```
DBI::dbListTables(con_air) # list tables in database
```

```
# access tables from the airlines database
```

```
flights <- dplyr::tbl(con_air, "flights")
```

```
carriers <- dplyr::tbl(con_air, "carriers")
```

```
airports <- dplyr::tbl(con_air, "airports")
```

```
DESCRIBE flights;
```

```
SELECT * FROM flights LIMIT 10;
```

```
DESCRIBE carriers;
```

```
SELECT * FROM carriers LIMIT 100;
```

```
DESCRIBE airports;
```

```
SELECT * FROM airports LIMIT 100;
```

3.2 Joins

In general, there are four pieces of information that you need to specify in order to join two tables:

- name of the first table
- type of join
- name of the second table
- condition(s) of the join

The following join types are available in MySQL dialect of SQL:

- **JOIN**
- **LEFT JOIN**
- **RIGHT JOIN**
- **CROSS JOIN**

```
-- access airport names for flights that flew out of Appleton in 2013
```

```
-- access both airport and carrier names for flights that flew out of Appleton in 2013
```

```
-- using LEFT JOIN
```

3.3 Unions

```
-- extract all flights that flew from ATW to MSP and from JFK to ORD on April 14, 2013
```

3.4 Subqueries

```
-- count the number of flights that satisfy the criteria above
```

```
# terminate the SQL connection
```

```
DBI::dbDisconnect(con_air, shutdown = TRUE)
```


4 Creating Databases

4.1 load packages and connect to database server

```
library(tidyverse)
library(mdsr)
library(DBI)
library(RMariaDB)
library(dbplyr)
library(duckdb)
```

As an example, we will consider the Saturday Night Live datasets available on the **snldb** GitHub [repo](#). Data is scraped from [SNL archives](#) and <http://www.imdb.com/title/tt0072562> by Hendrik Hilleckes and Colin Morris.

```
con_snl <- DBI::dbConnect(duckdb::duckdb(), dbdir = "snl_db")
```

```
SHOW TABLES;
```

4.2 accessing data (csv files) from GitHub (or, elsewhere)

Notice that there are eleven **.csv** files available in the **output** folder. Let's look at the **casts** dataset in R, so that we understand the data we want to load into the database.

```
# read in the data
```

```
casts <- readr::read_csv("")
```

```
glimpse(casts)
```

```
summary(casts)
```

```
# save the dataset as a .csv file on your (local) computer

write_csv(casts, "casts.csv")
```

4.3 data types in SQL

Consult [data types](#) for details.

4.4 importing data

Once the database is set up, you will be ready to import .csv files into the database as tables. Importing .csv files as tables requires a series of steps:

- a **USE** statement that ensures we are in the right database.
- a series of **DROP TABLE** statements that drop any old tables with the same names as the ones we are going to create.
- a series of **CREATE TABLE** statements that specify the table structures and field types.
- a series of **COPY** (or **LOAD DATA** for MySQL) statements that read the data from the .csv files into the appropriate tables.

```
USE snl_db;
```

```
DROP TABLE IF EXISTS casts;
```

```
CREATE TABLE casts (  
  aid VARCHAR(255) NOT NULL DEFAULT ' ',  
  sid SMALLINT NOT NULL DEFAULT 0,  
  featured BOOLEAN NOT NULL DEFAULT FALSE, /* use TINYINT(1) in MySQL instead of BOOLEAN) *,  
  first_epid INTEGER DEFAULT NULL,  
  last_epid INTEGER DEFAULT NULL,  
  update_anchor BOOLEAN NOT NULL DEFAULT FALSE,  
  n_episodes SMALLINT NOT NULL DEFAULT 0,  
  season_fraction DOUBLE NOT NULL DEFAULT 0,  
  PRIMARY KEY (sid, aid)  
);
```

```
COPY casts FROM 'casts.csv' (FORMAT CSV, HEADER, NULL 'NA');
```

4.5 checks

```
SHOW DATABASES;
```

```
SHOW TABLES;
```

```
DESCRIBE casts;
```

```
SELECT * FROM casts LIMIT 10;
```

4.6 shortcut (not accepted for Project 2)

```
# read in the data
```

```
episodes <- readr::read_csv("")
```

```
glimpse(episodes)
```

```
# save the dataset as a .csv file on your (local) computer
```

```
write_csv(episodes, "episodes.csv")
```

```
duckdb_read_csv(con = con_snl, name = "episodes", files = "episodes.csv")
```

```
SHOW TABLES;
```

```
DESCRIBE episodes;
```

```
SELECT * FROM episodes LIMIT 10;
```

```
# terminate the SQL connection  
DBI::dbDisconnect(con_snl, shutdown = TRUE)
```

Part II

Text Analysis

5 Case Study of Shakespeare's Macbeth using Regular Expressions

5.1 resources

- [Text Mining with R](#)
- [Modern Data Science with R \(3e\) - Chapter 19](#)

5.2 packages

```
library(tidyverse)
```

5.3 data

Our data comes from [Project Gutenberg](#), a large repository of free eBooks. We will be working with the text of Shakespeare's play Macbeth, which is available at <https://www.gutenberg.org/cache/epub/1129/pg1129.txt>. The text is in the public domain, so we are free to use it for our analysis.

```
library(gutenbergr)

data("gutenberg_metadata")

macbeth_all <- gutenberg_metadata |>
  filter(str_detect(title, "Macbeth"))

book <- gutenberg_download(1129, meta_fields = "title")
```

For our case study, we will be working with the raw text of the play, which is available in the `mdsr` package.

```
library(mdsr)

data(Macbeth_raw)

?Macbeth_raw
```

```
macbeth <- Macbeth_raw |>
  str_split("\r\n") |>
  pluck(1)

length(macbeth)

macbeth[300:310]
```

5.4 strings and regular expressions

```
# how many times does the character Macbeth speak in the play?

macbeth_lines <- macbeth |>
  str_subset(" MACBETH")

length(macbeth_lines)

head(macbeth_lines)
```

```
macbeth |>
  str_which(" MACBETH")
```

5.5 about regular expressions grammar

5.5.1 metacharacter

```
macbeth |>
  str_subset("MAC.") |>
  head()
```

```
macbeth |>
  str_subset("MACBETH\\.") |>
  head()
```

5.5.2 character sets

```
macbeth |>
  str_subset("MAC[B-Z]") |>
  head()
```

5.5.3 alternation

```
macbeth |>
  str_subset("MAC(B|D)") |>
  head()
```

5.5.4 anchors (power-dollar)

```
macbeth |>
  str_subset("^ MAC[B-Z]") |>
  head()
```

```
macbeth |>
  str_subset("\\\\.$") |>
  head()
```

5.5.5 repetitions

```
# ?

macbeth |>
  str_subset("^ ?MAC[B-Z]") |>
  head()
```



```
# *

macbeth |>
  str_subset("^ *MAC[B-Z]") |>
  head()
```

```
# +

macbeth |>
  str_subset("^ +MAC[B-Z]") |>
  head()
```

5.6 a case study - life and death in Macbeth and speaking patterns

```
macbeth_chars <- tribble(
  ~name, ~regexp,
  "Macbeth", "  MACBETH\\.",
  "Lady Macbeth", "  LADY MACBETH\\.",
  "Banquo", "  BANQUO\\.",
  "Duncan", "  DUNCAN\\.",
) |>
  mutate(speaks = purrr::map(regexp, str_detect, string = macbeth))
```

```
speaker_freq <- macbeth_chars |>
  unnest(cols = speaks) |>
  mutate(
    line = rep(1:length(macbeth), 4),
    speaks = as.numeric(speaks)
  ) |>
  filter(line > 218 & line < 3172)

glimpse(speaker_freq)
```

```
acts <- tibble(
  line = str_which(macbeth, "^ACT [I|V]+"),
  line_text = str_subset(macbeth, "^ACT [I|V]+"),
  labels = str_extract(line_text, "^ACT [I|V]+")
)
```

```

ggplot(data = speaker_freq, aes(x = line, y = speaks)) +
  geom_smooth(
    aes(color = name), method = "loess",
    se = FALSE, span = 0.4
  ) +
  geom_vline(
    data = acts,
    aes(xintercept = line),
    color = "darkgray", lty = 3
  ) +
  geom_text(
    data = acts,
    aes(y = 0.085, label = labels),
    hjust = "left", color = "darkgray"
  ) +
  ylim(c(0, NA)) +
  xlab("Line Number") +
  ylab("Proportion of Speeches") +
  scale_color_colorblind()

```

6 Case Study of Donald Trump's tweets during the 2016 US presidential election

6.1 resources

- [Trump Twitter Archive](#)
- [Todd Vaziri's Tweet](#)
- [David Robinson's Blog Post](#)
- [Text Mining With R](#)

6.2 packages

```
library(tidyverse)
library(scales)
library(tidytext)
library(textdata)
library(dslabs)
```

6.3 data

Our data come from the *dslabs* package.

```
data("trump_tweets")

?trump_tweets

glimpse(trump_tweets)
```

```
range(trump_tweets$created_at)
```

```
trump_tweets$text[16413]
```

```
trump_tweets |>  
  count(source) |>  
  arrange(desc(n))
```

For our case study, we are interested in what happened during the 2016 campaign, so for this analysis we will focus on what was tweeted between the day Trump announced his campaign and election day. Also, we only keep tweets from Android or iPhone, and filter out retweets.

```
campaign_tweets <- trump_tweets |>  
  filter(source %in% paste("Twitter for", c("Android", "iPhone"))) &  
    created_at >= ymd("2015-06-17") &  
    created_at < ymd("2016-11-08")) |>  
  mutate(source = str_remove(source, pattern = "Twitter for ")) |>  
  filter(!is_retweet) |>  
  arrange(created_at) |>  
  as_tibble()
```

```
campaign_tweets |>  
  count(source)
```

6.4 some initial exploration

We explore the possibility that two different groups were tweeting from these devices.

```
campaign_tweets |>  
  mutate(hour = hour(with_tz(created_at, "EST"))) |>  
  count(source, hour)
```

```
android_vs_iphone_hourly_props <- campaign_tweets |>  
  mutate(hour = hour(with_tz(created_at, "EST"))) |>  
  count(source, hour) |>  
  group_by(source) |>  
  mutate(percent = n / sum(n))  
  
android_vs_iphone_hourly_props
```

```
android_vs_iphone_hourly_props |>
  ggplot(aes(x = hour, y = percent, color = source)) +
  geom_line() +
  geom_point() +
  scale_y_continuous(labels = percent_format()) +
  labs(x = "Hour of day (EST)", y = "% of tweets", color = "")
```

6.5 tokens

```
i <- 3008

campaign_tweets$text[i]
```

```
campaign_tweets[i,] |>
  tidytext::unnest_tokens(output = word, input = text, token = "words") |>
  dplyr::pull(word)
```

```
links_to_pics <- "(https://t.co/[A-Za-z\\d]+)|(http://t.co/[A-Za-z\\d]+)"

campaign_tweets[i,] |>
  mutate(text = str_remove_all(text, pattern = links_to_pics)) |>
  tidytext::unnest_tokens(output = word, input = text, token = "words") |>
  dplyr::pull(word)
```

```
tweet_words <- campaign_tweets |>
  mutate(text = str_remove_all(text, pattern = links_to_pics)) |>
  tidytext::unnest_tokens(output = word, input = text, token = "words")
```

```
tweet_words |>
  count(word) |>
  arrange(desc(n))
```

6.6 stop words

```
data(stop_words)
```

```
?stop_words
```

```
tweet_words <- campaign_tweets |>
  mutate(text = str_remove_all(text, pattern = links_to_pics)) |>
  tidytext::unnest_tokens(output = word, input = text, token = "words") |>
  filter(!(word %in% stop_words$word))
```

```
tweet_words |>
  count(word)
```

```
tweet_words <- campaign_tweets |>
  mutate(text = str_remove_all(text, links_to_pics)) |>
  tidytext::unnest_tokens(output = word, input = text, token = "words") |>
  filter(!(word %in% stop_words$word)) |>
  filter(!(str_detect(word, pattern = "^\\d+")))
```

```
tweet_words |>
  count(word)
```

```
# |>
#   arrange(desc(n))
```

6.7 further exploration

```
android_vs_iphone <- tweet_words |>
  count(word, source)
```

```
android_vs_iphone
```

```
android_vs_iphone <- tweet_words |>
  count(word, source) |>
  pivot_wider(names_from = "source", values_from = "n", values_fill = 0)
```

```
android_vs_iphone
```

```

android_vs_iphone <- tweet_words |>
  count(word, source) |>
  pivot_wider(names_from = "source", values_from = "n", values_fill = 0) |>
  mutate(p_a = (Android + 0.5)/(sum(Android) + 0.5),
         p_i = (iPhone + 0.5)/(sum(iPhone) + 0.5),
         ratio = p_a / p_i)

android_vs_iphone

```

for words appearing at least 100 times in total, here are the ten highest percent differences

```

top_10_android <- android_vs_iphone |>
  filter(Android + iPhone >= 100) |>
  arrange(desc(ratio)) |>
  slice_head(n = 10)

top_10_android

```

and the top ten for iPhone

```

top_10_iphone <- android_vs_iphone |>
  filter(Android + iPhone >= 100) |>
  arrange(ratio) |>
  slice_head(n = 10)

top_10_iphone

```

```

top_10_android |>
  rbind(top_10_iphone) |>
  dplyr::select(word, iPhone, Android, ratio) |>
  mutate(log_ratio = log(ratio)) |>
  mutate(device = rep(c("Android", "iPhone"), each = 10)) |>
  ggplot() +
  geom_col(aes(x = log_ratio, y = fct_reorder(word, log_ratio), fill = device))

```

6.8 sentiment analysis

```

tidytext::get_sentiments("bing")

```

```
tidytext::get_sentiments("afinn")
```

```
tidytext::get_sentiments("nrc")
```

```
nrc <- tidytext::get_sentiments("nrc")
```

```
nrc |> count(sentiment)
```

```
tweet_words |>  
  left_join(nrc, by = "word") |>  
  dplyr::select(source, word, sentiment) |>  
  sample_n(5)
```

```
sentiment_counts <- tweet_words |>  
  left_join(nrc, by = "word", relationship = "many-to-many") |>  
  count(source, sentiment)
```

```
sentiment_counts
```

```
sentiment_counts <- tweet_words |>  
  left_join(nrc, by = "word", relationship = "many-to-many") |>  
  count(source, sentiment) |>  
  pivot_wider(names_from = "source", values_from = "n") |>  
  mutate(sentiment = replace_na(sentiment, replace = "none"))
```

```
sentiment_counts
```

```
sentiment_counts <- sentiment_counts |>  
  mutate(p_a = Android/sum(Android),  
         p_i = iPhone/sum(iPhone),  
         ratio = p_a/p_i) |>  
  arrange(desc(ratio))
```

```
sentiment_counts
```

```
ordered_levels <- sentiment_counts$sentiment
```

```
android_vs_iphone |>  
  filter(Android + iPhone >= 10) |>  
  left_join(nrc, by = "word") |>
```



```
mutate(sentiment = factor(sentiment, levels = sentiment_counts$sentiment)) |>
group_by(sentiment) |>
slice_max(abs(log(ratio)), n = 10) |>
ungroup() |>
mutate(word = reorder(word, -ratio)) |>
ggplot(aes(word, ratio, fill = ratio < 1)) +
geom_col(show.legend = FALSE) +
scale_y_log10() +
facet_wrap(~sentiment, scales = "free_x", nrow = 2) +
theme(axis.text.x = element_text(angle = 90, hjust = 1))
```

7 Case Study of Data Science Papers from arXiv.org

7.1 resources

- [arXiv.org](https://arxiv.org)

7.2 packages

```
library(tidyverse)
library(scales)
library(tidytext)
library(textdata)
library(mdsr)
```

7.3 data

We can import data using the **aRxiv** package.

```
library(aRxiv)

df <- arxiv_search(
  query = 'ti:"Data Science"',
  limit = 20000,
  batchsize = 100
)

glimpse(df)
```

For this case study we will use some already-collected data from the **mdsr** package.

```
library(mdsr)

data(DataSciencePapers)

?DataSciencePapers

glimpse(DataSciencePapers)
```

7.4 initial data exploration and cleaning

```
DataSciencePapers <- DataSciencePapers |>
  mutate(
    submitted = lubridate::ymd_hms(submitted),
    updated = lubridate::ymd_hms(updated)
  )

glimpse(DataSciencePapers)
```

```
mosaic::tally(~ year(submitted), data = DataSciencePapers)
```

```
DataSciencePapers |>
  filter(id == "1809.02408v2") |>
  glimpse()
```

```
DataSciencePapers |>
  group_by(primary_category) |>
  count()
```

```
DataSciencePapers <- DataSciencePapers |>
  mutate(field = str_extract(primary_category, pattern = "[a-z,-]+"))

mosaic::tally(x = ~field, margins = TRUE, data = DataSciencePapers) |>
  sort()
```

7.5 corpora and tokenization

```
DataSciencePapers |>
  unnest_tokens(output = word, input = abstract, token = "words") |>
  count(id, word, sort = TRUE)
```

```
arxiv_words <- DataSciencePapers |>
  unnest_tokens(output = word, input = abstract, token = "words") |>
  anti_join(tidytext::get_stopwords(), by = "word")

arxiv_words |>
  count(id, word, sort = TRUE)
```

```
arxiv_abstracts <- arxiv_words |>
  group_by(id) |>
  summarize(abstract_clean = paste(word, collapse = " "))

arxiv_papers <- DataSciencePapers |>
  left_join(arxiv_abstracts, by = "id")
```

7.6 wordcloud

```
library(wordcloud)

set.seed(4302026)
arxiv_papers |>
  pull(abstract_clean) |>
  wordcloud(
    max.words = 40,
    scale = c(8, 1),
    colors = topo.colors(n = 30),
    random.color = TRUE
  )
```

7.7 bigrams

```
arxiv_bigrams <- arxiv_papers |>
  unnest_tokens(
```

```

    output = arxiv_bigram,
    input = abstract_clean,
    token = "ngrams",
    n = 2
  ) |>
  dplyr::select(arxiv_bigram, id)

arxiv_bigrams

arxiv_bigrams |>
  count(arxiv_bigram, sort = TRUE)

```

7.8 tf-idf and document term matrices

```

arxiv_words |>
  count(word) |>
  arrange(desc(n)) |>
  head()

```

```

tidy_DTM <- arxiv_words |>
  count(id, word) |>
  tidytext::bind_tf_idf(word, id, n)

tidy_DTM

```

```

tidy_DTM |>
  arrange(desc(tf)) |>
  head()

```

```

tidy_DTM |>
  arrange(desc(idf), desc(n)) |>
  head()

```

```

tidy_DTM |>
  filter(word == "wildfire")

```

```

arxiv_papers |>
  pull(abstract) |>

```

```
str_subset("wildfire") |>  
strwrap() |>  
head()
```

```
tidy_DTM |>  
  filter(word == "implications")
```

```
tidy_DTM |>  
  filter(id == "1809.02408v2") |>  
  arrange(desc(tf_idf)) |>  
  head()
```

```
tm_DTM <- arxiv_words |>  
  count(id, word) |>  
  tidytext::cast_dtm(document = id, term = word, value = n)  
  
tm_DTM
```

Part III

Geospatial Data

8 Recreating John Snow's 1854 map of the Broad Street cholera outbreak.

8.1 resources

- [Modern Data Science with R \(3e\) - Chapter 17](#)
- [1854 Broad Street cholera outbreak](#)
- [Shapefile](#)

8.2 packages

```
library(tidyverse)
library(sf)
library(ggspatial)
library(maps)
library(mapproj)
```

8.3 data

```
root <- here::here("/cloud/project") # where we stored these files
dsn <- fs::path(root, "SnowGIS_SHP")
```

```
list.files(dsn)
```

```
st_layers(dsn)
```



```
CholeraDeaths <- st_read(dsn, layer = "Cholera_Deaths")
```

```
class(CholeraDeaths)
```

```
plot(CholeraDeaths["Count"])
```

```
ggplot(CholeraDeaths) +  
  geom_sf()
```

```
ggplot(CholeraDeaths) +  
  ggspatial::annotation_map_tile(type = "osm", zoomin = 0, progress = "none") +  
  geom_sf(aes(size = Count), alpha = 0.7)
```

```
# bounding boxes
```

```
st_bbox(CholeraDeaths)
```

8.4 projections

[Top 10 World Map Projections](#)

```
maps::map(database = "world", projection = "mercator", wrap = TRUE)
```

```
maps::map(database = "world", projection = "cylequalarea", param = 45, wrap = TRUE)
```

8.5 CRS - coordinate reference system

[Coordinate Reference System](#) - three formats that are common for storing information about the projection of a geospatial object are EPSG, PROJ.4, and WKT

Common CRSs

- [EPSG:4326](#) - also known as WGS84, standard for GPS systems and Google Earth
- [EPSG:3857](#) - a Mercator projection used in map tiles by Google Maps, Open Street Maps, etc.
- [EPSG:27700](#) - also known as OSGB 1936, or the British National Grid, commonly used in Britain

```
st_crs(4326)$epsg  
  
st_crs(3857)$epsg  
  
st_crs(27700)$epsg
```

```
st_crs(4326)$Wkt  
  
st_crs(3857)$Wkt  
  
st_crs(27700)$Wkt
```

```
st_crs(4326)$proj4string  
  
st_crs(3857)$proj4string  
  
st_crs(27700)$proj4string
```

8.6 what about our dataset?

```
st_crs(CholeraDeaths)
```

```
cholera_4326 <- CholeraDeaths |>  
  st_transform(4326)  
  
st_bbox(cholera_4326)
```

```
ggplot(cholera_4326) +  
  annotation_map_tile(type = "osm", zoomin = 0) +  
  geom_sf(aes(size = Count), alpha = 0.7)
```

```
# add pump locations  
  
pumps <- st_read(dsn, layer = "Pumps")  
  
pumps_4326 <- pumps |>  
  st_transform(4326)
```

```
ggplot(cholera_4326) +  
  annotation_map_tile(type = "osm", zoomin = 0) +  
  geom_sf(aes(size = Count), alpha = 0.7) +  
  geom_sf(data = pumps_4326, color = "red", size = 3)
```

Part IV

Code Performance

9 Code Performance

9.1 resources

- [Advanced R](#) - Chapters 22-25

9.2 code profiling

```
# load the package
library(profvis) # for profiling R code and visualizing the results
```

```
# example 1

f <- function() {
  pause(0.1)
  g()
  h()
}

g <- function() {
  pause(0.1)
  h()
}

h <- function() {
  pause(0.1)
}

####

profvis(f())
```

```
# example 2

# a slow, iterative random walk in R
random_walk_r <- function(n) {
  x <- numeric(n)
  x[1] <- 0
  for(i in 2:n) {
    x[i] <- x[i-1] + rnorm(1, mean = 0, sd = 1)
  }
  return(x)
}

####

profvis::profvis({
  # running a long simulation to give the profiler enough data
  sim_data <- random_walk_r(1e5)
})
```

9.3 improving performance with Rcpp

```
# load the package
library(Rcpp) # for integrating C++ code into R

# define the C++ function as a string (or in a separate .cpp file)

Rcpp::cppFunction('
  NumericVector random_walk_cpp(int n) {
    NumericVector x(n);
    x[0] = 0;
    for(int i = 1; i < n; ++i) {
      x[i] = x[i-1] + R::rnorm(0, 1);
    }
    return x;
  }
')
```

OR see below

```
#include <Rcpp.h>
using namespace Rcpp;

// [[Rcpp::export]]

NumericVector random_walk_cpp_2(int n) {
  NumericVector x(n);
  x[0] = 0;
  for(int i = 1; i < n; ++i) {
    x[i] = x[i-1] + R::rnorm(0, 1);
  }
  return x;
}
```

```
profvis::profvis({
  # running a long simulation to give the profiler enough data
  sim_data_cpp <- random_walk_cpp(1e5)
})
```

```
# install.packages("bench")
library(bench)

# compare the R vs C++ implementations
results <- bench::mark(
  r_version = random_walk_r(10000),
  cpp_version = random_walk_cpp(10000),
  iterations = 100,
  check = FALSE # don't check for exact identical output due to random seeds
)
```

```
results |>
  dplyr::select(expression, min, median, `itr/sec`, mem_alloc)
```

```
plot(results)
```

9.4 parallelization with future and furr

```
# load the packages
library(tictoc) # for timing our code execution
```

```
library(future)    # the core parallel backend architecture
library(furrr)     # a parallel implementation of the purrr family of functions
```

```
future::availableCores()
```

```
future::plan(multisession, workers = 2)
```

```
simulate_median <- function(iteration, n = 1000) {
  # simulate heavily skewed data (e.g., Log-Normal)
  sample_data <- rlnorm(n, meanlog = 10, sdlog = 1)

  # return a tibble with the result
  tibble(
    iteration = iteration,
    median_val = median(sample_data)
  )
}
```

```
# always test the function at least once before parallelizing
```

```
simulate_median(iteration = 1)
```

```
tic("Sequential Purrr")
set.seed(405)
results_seq <- 1:1e4 |>
  purrr::map_dfr(.f = simulate_median)
toc()
```

```
tic("Parallel Furrr")
set.seed(405)
results_par <- 1:1e4 |>
  furrr::future_map_dfr(.f = simulate_median, .options = furrr_options(seed = TRUE))
toc()
```

9.5 efficient file handling

```
# load the packages
library(tidyverse)    # for data manipulation and writing CSV files
library(nanoparquet)  # for writing Parquet files in R
```



```
write_csv(results_seq, "results_seq.csv")

write_parquet(results_seq, "results_seq.parquet")

read_bench <- bench::mark(
  csv = read_csv("results_seq.csv", show_col_types = FALSE),
  parquet = read_parquet("results_seq.parquet"),
  iterations = 15,
  check = FALSE
)

read_bench |> select(expression, median)

plot(read_bench)
```

Part V

R Package

10 Building an R Package

10.1 check for available package names

```
# check available package names (chapter 4.1.2 of R Packages (2e))
install.packages(c("available", "pak"))
available("package_name")
pak::pkg_name_check("package_name")
```

10.2 create package

```
# start by creating a GitHub repo with the package name
# clone that repo to your Posit Cloud
```

```
# install some packages which are required for package development
install.packages(c("devtools", "checkmate", "tidyverse", "remotes"))
# devtools - meta package for package development in R
# checkmate - to build assertions (make sure that our functions are working fine)
# tidyverse - to build one of our functions
# remotes is a package that is required to install a package
```

```
library(devtools)
```

```
# create the package files (basic)
usethis::create_package("/cloud/project")
# make sure to hit the 'Yes' option and check the files/folders that have been created
# add your package name to the Package field in the DESCRIPTION file
# commit and push to GitHub
```

```
library(devtools)
```

10.3 create functions within the package

```
# existing function called strsplit that comes with base R
strsplit("alpha, bravo", split = ",")
# this returns by default a list which is annoying
strsplit("alpha, bravo", split = ",")[[1]]
# we will create a new function called strsplit1 that returns a vector instead of a list
```

```
# make sure the directory is okay
getwd() # should see /cloud/project
```

```
# every function in an R package needs to be written in an .R script with the same name
# which will be stored in the R directory
usethis::use_r("strsplit1")
# type in function R script
strsplit1 <- function(x, split)
{
  strsplit(x, split = split)[[1]]
}
```

```
devtools::load_all()
# loads the package from source directly into the current environment
# temporarily builds and installs the package, allows us to verify our functions
```

```
# check if your functions works
strsplit1("alpha, bravo", split = ",")
# commit and push to GitHub
```

```
# create the min_max_scale function
usethis::use_r("min_max_scale")
# type in function R script
min_max_scale <- function(x)
{
  (x - min(x, na.rm = TRUE)) / (max(x, na.rm = TRUE) - min(x, na.rm = TRUE))
}

devtools::load_all()

# check
min_max_scale(1:5)
# commit and push to GitHub
```

```

# create a personal theme (theme_academic)
usethis::use_r("theme_academic")
# type in function R script
theme_academic <- function(base_size = 12) {
  ggplot2::theme_classic(base_size = base_size) +
    ggplot2::theme(
      plot.title = ggplot2::element_text(face = "bold", hjust = 0.5),
      panel.grid.major = ggplot2::element_blank(),
      panel.grid.minor = ggplot2::element_blank()
    )
}
# specifically mention what external packages are used to build functions in your package
usethis::use_package("ggplot2")
# check Imports field in DESCRIPTION file

devtools::load_all()

# check
library(ggplot2)
p <- ggplot(mtcars) + geom_point(aes(x = mpg, y = hp)) + labs(title = "plot title")
p
p + theme_academic()
# commit and push to GitHub

```

```

# so we have created the barebones version of all our three functions

```

```

# we updated the DESCRIPTION file (changes to title, authors, description)
# most of the other things will get updated on their own
devtools::check()

# run the above check almost after every step
# should get 0 errors, 0 warnings, 0 notes
# right now we get a warning/note about license

```

10.4 add license

```

# let's add a license
usethis::use_mit_license()
# we won't change the LICENSE and LICENSE.md file by hand

```

```
devtools::check() # check again
# commit and push to GitHub
```

10.5 add documentations

```
# we are now going add documentations to our functions
# go to the function .R script > Code> Insert roxygen skeleton
# make sure to have the @export tag
# think of a user when you are writing documentations
devtools::document() # this needs to be run everytime we update the documentation
devtools::load_all()
devtools::check()
# commit and push to GitHub
```

10.6 add assertions

```
# we will work with assertions from the checkmate package
# first check of the function arguments so that it runs smoothly
# checkmate provides somewhat better messages
usethis::use_package("checkmate")
# we will write assertions for each input of each function
# see help pages of ?assert_
devtools::load_all()
# add assertions to each function and run devtools::load_all() after each change
# tryto break your function and see if the assertion works
devtools::document()
devtools::check()
# commit and push to GitHub
```

10.7 unit testing

```
# unit testing of functions
# think about every possible sceanrio where your function can break
usethis::use_testthat()
```

```

usethis::use_test("strsplit1")
# generate a new file inside the testthat directory inside tests directory
# we will delete the prepopulated message and write our own for our functions
# see help pages of ?expect_
# write your unit tests
devtools::load_all()
devtools::test()

```

```

usethis::use_test("min_max_scale")
devtools::load_all()
devtools::test()

```

```

usethis::use_test("theme_academic")
devtools::load_all()
devtools::test()

```

```

devtools::check()
# commit and push to GitHub

```

10.8 add dataset

```

# add dataset to our package that can be exported and used by our users
# first make sure that the dataset is created in your local Environment
toy_data <- data.frame(
  subject_id = 1:5,
  reaction_time = c(1.2, 0.9, 1.5, 0.8, 1.1),
  test_score = c(45, 82, 55, 91, 67),
  recalled_words = c(
    "cat, dog, mouse",
    "apple, banana",
    "car, truck, bike, train",
    "red, blue, green",
    "sun, moon"
  )
)

```

```

usethis::use_data(toy_data)
# if exporting big datasets, look into the compress argument of use_this

```

```
# we will save the data generating script into a file (for reproducibility)
# very important if you generate datasets that need random generation (set a seed)
usethis::use_data_raw("toy_data_code")
# anytime you update the toy_data with new code, make sure to update the toy_data_code file
```

```
# documentation for toy_data
usethis::use_r("toy_data")
# seems to be done manually
# this will build the documentation
# make sure that the last line has the name of the dataset inside ""
# mention the units of your variables
```

```
# if we export multiple datasets, R puts them all in sysdata.rda file
# in the NAMESPACE, it doesn't specifically mention the dataset to be exported
```

```
devtools::document()
devtools::load_all() # display the dataset and check help page
devtools::check()
# commit and push to GitHub
```

```
# for any file that is typically not an rda file, like csv files
# we will need to create a directory called inst/extdata
```

10.9 add README

```
# we will build the README for the GitHub site and also for our website
usethis::use_readme_rmd()
# try not to install any other packages (since those packages have not been mentioned in Imports)
devtools::build_readme()
# use this build function repeatedly whenever you update the README.Rmd
devtools::check()
# commit and push to GitHub
# check GitHub
```

10.10 add/build vignettes


```
# build a vignette
# these are specific instructions/notes about very specific part of your package
# for example check out the ggplot2 vignettes from Articles dropdown of the package website
usethis::use_vignette(name = "intro-to-toypackage", title = "Introduction to toypackage")
devtools::build_vignettes() # to convert the Rmd into website readable html
devtools::check()
# commit and push to GitHub
```

10.11 build package website

```
# build the package website
# pkgdown package this is how the website will be built
usethis::use_pkgdown() # run this only once
# the website will also use GitHub pages like it did for our personal website
pkgdown::build_site() # run this everytime there is an update to anything inside your package
usethis::use_pkgdown_github_pages() # notice the changes, run this only once
devtools::check()
# commit and push to GitHub
```